

Tutorial 8

Question 9.2

Using the wave:

$$i_s = 15 \cos(25\pi t + 25^\circ) A$$

Find:

- | | |
|-----------|-----------------------|
| 01 | Amplitude [1] |
| 02 | Angular frequency [1] |
| 03 | Frequency [1] |
| 04 | Current at 2 ms [1] |

Question 9.6

Find which wave leads and give the magnitude.

05) $v(t) = 10 \cos(4t - 60^\circ)V$

$i(t) = 4 \sin(4t + 50^\circ)A$

06) $v_1(t) = 4 \cos(377t + 10^\circ)V$

$v_2(t) = -20 \cos(377t)V$

07) $x(t) = 13 \cos(2t) + 5 \sin(2t)$

$y(t) = 15 \sin(2t - 11.8^\circ)$

Question 9.13	
Simplify the following phasors.	
08)	$\frac{2 + j3}{1 - j6} + \frac{7 - j8}{-5 + j11}$
09)	$\frac{(5\angle 10^\circ)(10\angle -40^\circ)}{(4\angle -80^\circ)(-6\angle 50^\circ)}$
10)	$\begin{vmatrix} 2 + j3 & -j2 \\ -j2 & 8 - j5 \end{vmatrix}$

Question 9.19	
Simplify the following:	
11)	$3 \cos (20t + 10^\circ) - 5 \cos (20t - 30^\circ)$
12)	$40 \sin (50t) + 30 \cos (50t - 45^\circ)$
13)	$20 \sin (400t) + 10 \cos (400t + 60^\circ) - 5 \sin (400t - 20^\circ)$

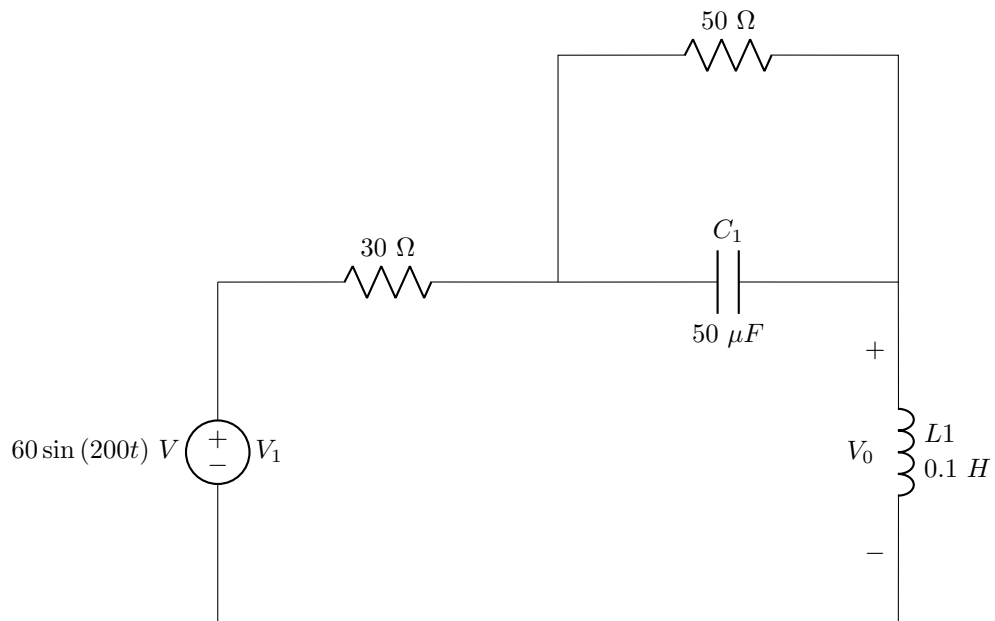
Question 9.20

Find **Z** if:

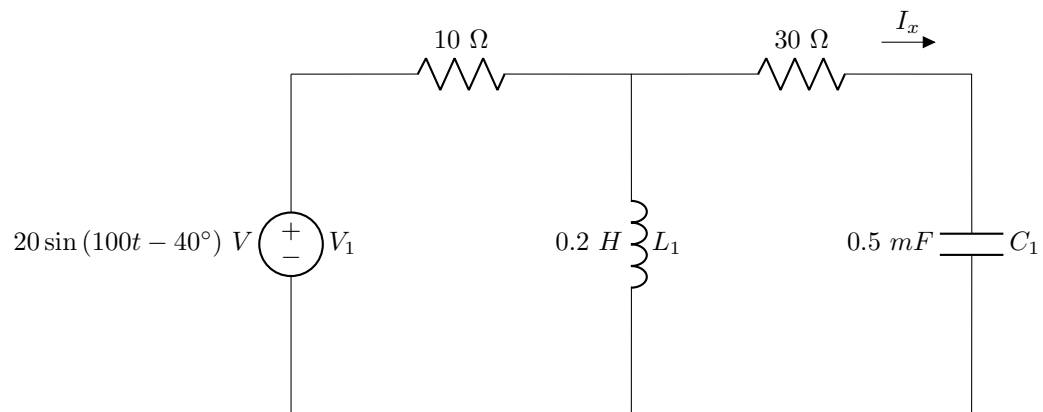
$$i = 7.5 \cos(10t + 30^\circ)$$

$$v = 120 \cos(10t + 75^\circ)$$

Question 9.42

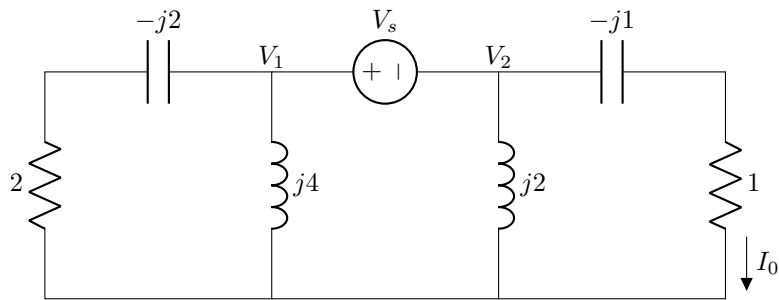
Find V_0 :

Question 9.48

Find $\mathbf{I}_x$:

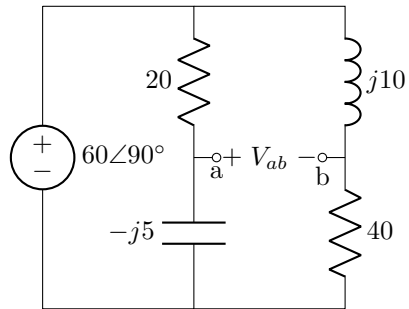
Question 9.54

Find $\mathbf{V}_s$ if $\mathbf{I}_0 = 2 \text{ A}$:



Question 9.66

Find $\mathbf{Z}_T$ and $\mathbf{V}_{ab}$:



Question 9.70

Find $\mathbf{Z}_{eq}$: